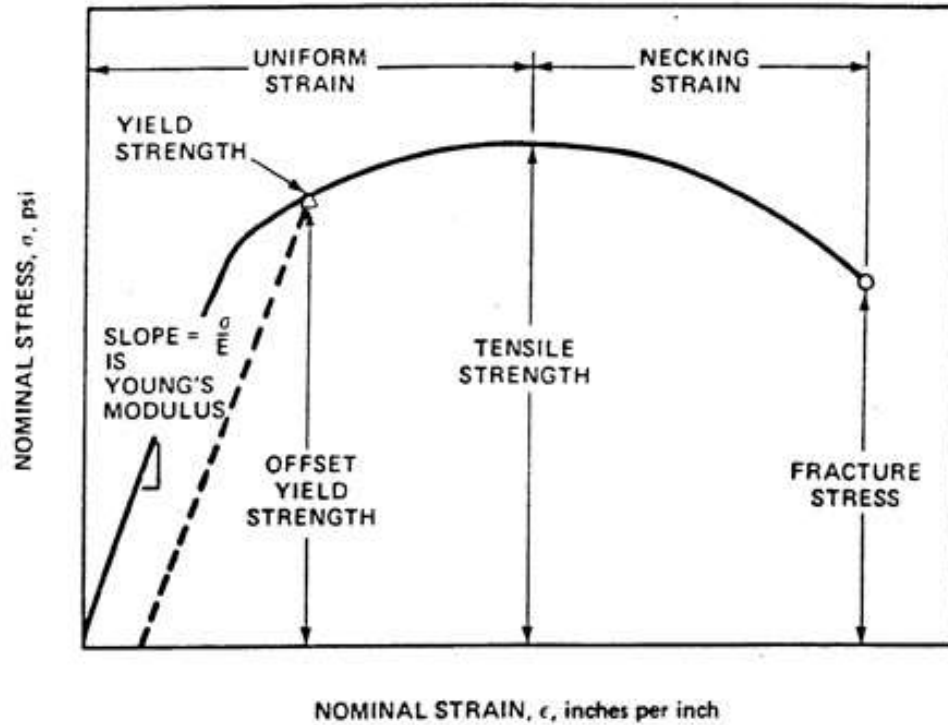


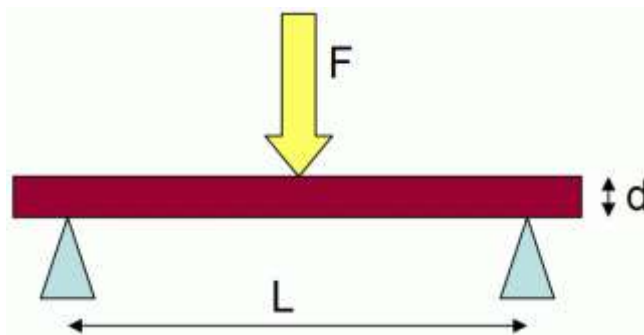
STRESS-STRAIN BEHAVIOR



Flexural strength (3-point bending test)

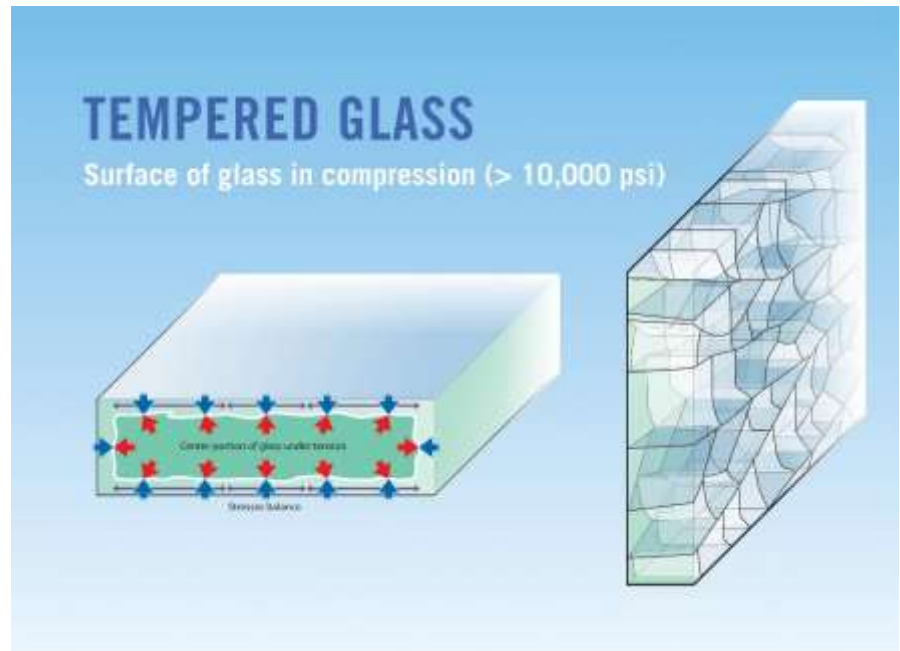
Better test for ceramic compared to tensile test

(We can't use a tensile test to determine the strength of ceramics, because it is difficult to prepare/position specimens accurately. Specimens often break in the grips and ceramics break after 0.1% strain, meaning that the specimen must be perfectly aligned to avoid any bending stresses)

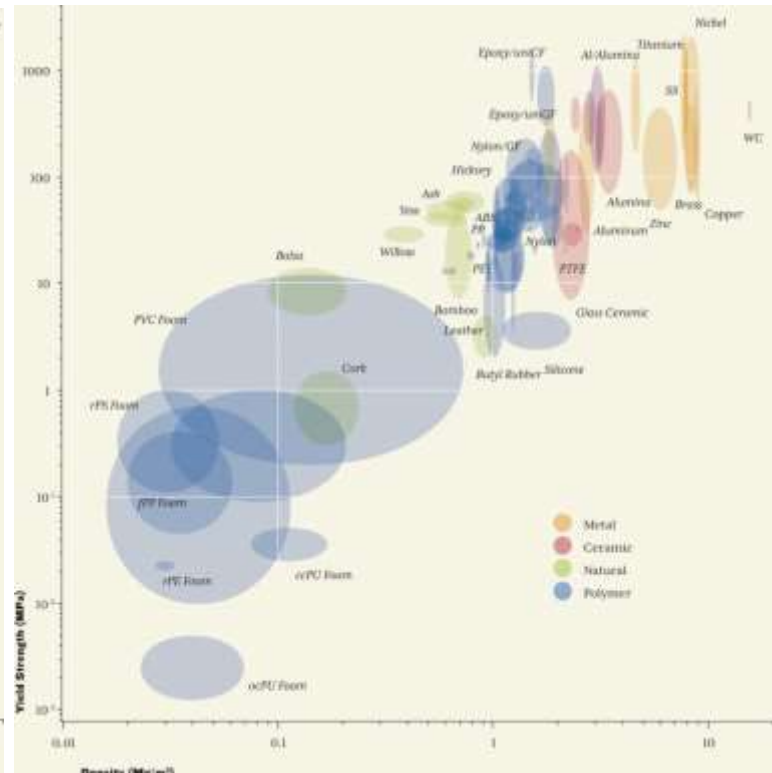
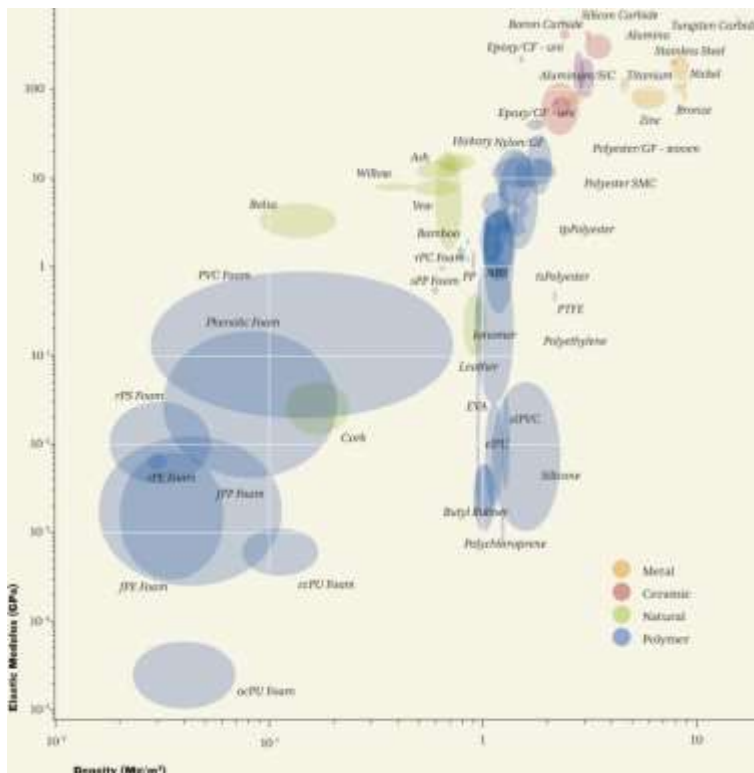


$$\sigma = \frac{3Fl}{2bd^2}$$

This stress causes the glass to crumble into small granular chunks when broken (safer)



(Focus on the relative position of the different materials)



Learning objectives of the week (accessible in the syllabus section in Quercus)

- Justify the statement that a scientific model need only be as good as it needs to be for the present discussion
- Using two examples from the fields of materials science or solid state chemistry, demonstrate the usefulness and the limitations of a scientific model
- Contrast plastic deformation with elastic deformation
- Provide at least one atomic level definition of plastic deformation and one macroscopic definition
- Explain why the term "plastic" is used to refer to permanent deformation
- Sketch generalized curves for the stress-strain behaviour of metals, ceramics, and polymers
- Contrast the mechanical behaviour of ceramics with that of metals and polymers
- Summarize the need to test ceramics in bending, rather than in a tensile test
- Given all of the others, calculate any one of the following for a 3-point bend test on a rectangular sample: sample height and width, span between lower supports, load, peak stress in sample
- Describe the general theory behind tempered glass, whether thermally or chemically tempered
- Illustrate the final stress distribution through the thickness of a sheet of tempered glass
- Explain how residual stresses increase the strength and improve the safety of tempered glass
- Generalize the relative positions on a logarithmic plot of strength versus density and Young's modulus versus density of several metals, ceramics, and polymers
- Identify the region on a plot of Young's modulus versus density where we would like to develop new materials for many applications such as aerospace
- OPTIONAL: Demonstrate, given the derivation, how the Materials Performance Index for a light stiff beam is determined to be $E^{1/2}/\rho$

1. Which of the following statements are **TRUE**?

- A. Polymers are generally more ductile than metals, which (metals) are more ductile than ceramics.
- B. Metals are generally stronger than polymers, which (polymers) are stronger than ceramics.
- C. Young's modulus is structural independent.

- a) A and B
- b) B only
- c) A and C
- d) C only

2. The Young's modulus describes which of the following types of deformation?

- a) Uniform elastic
- b) Non-uniform (localized) elastic
- c) Uniform Plastic
- d) Non-uniform (localized) plastic

Answer : 1.c) , 2.a)

Question 1: (6 Marks)

Determine what material fits the description (show your work)

- i) The maximum length that can be reached by applying tension on a 50 cm metal bar before it begins to plastically deform is 50.04 cm. (2 Marks)
- ii) The material with the lowest strain at yield strength. (1 Mark)
- iii) The materials with the highest strain at yield strength (1 Mark)
- iv) The required applied force to plastically deform a tensile specimen with a rectangular cross-sectional area (10 mm x 5 mm) is 22.5 kN. (2 Marks)

<i>Material</i>	<i>Yield Strength [MPa]</i>	<i>Modulus of Elasticity [GPa]</i>	<i>Strain- Hardening exponent</i>	<i>K MPa</i>
<i>Tungsten</i>	<i>550</i>	<i>407</i>	<i>--</i>	<i>--</i>
<i>Steel</i>	<i>180</i>	<i>207</i>	<i>0.21</i>	<i>600</i>
<i>Nickel</i>	<i>138</i>	<i>190</i>	<i>--</i>	<i>--</i>
<i>Titanium</i>	<i>450</i>	<i>107</i>	<i>0.15</i>	<i>1400</i>
<i>Brass</i>	<i>77</i>	<i>97</i>	<i>0.20</i>	<i>585</i>
<i>Aluminum</i>	<i>35</i>	<i>69</i>	<i>0.17</i>	<i>780</i>
<i>Magnesium</i>	<i>130</i>	<i>45</i>	<i>0.16</i>	<i>450</i>

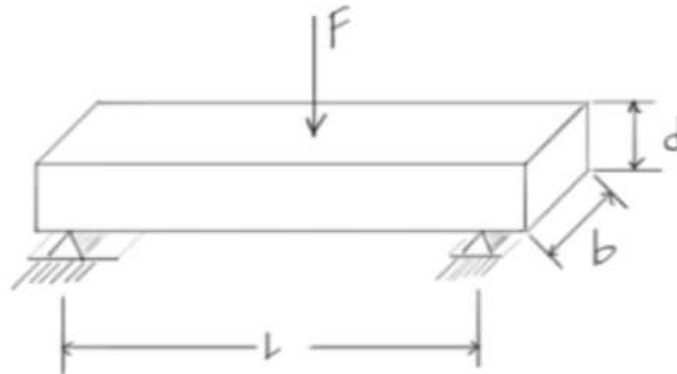
Answer :

Question 1:

- i. Specimen is made out of Brass
- ii. Aluminum has the lowest strain at yield stress
- iii. Titanium has highest strain at yield stress
- iv. Stress = 450 MPa. Specimen is made out of Titanium

Part B. Long Answers – Show all your works to obtain full mark.

6. A bar of 301 steel is loaded by 960 N in three-point bending, as shown in the figure below. The bar has a length of 70 cm, a width of 4 cm and a height of 1.5 cm. This sample has a yield strength of 515 MPa. Will this piece of specimen plastically deform?



$$\sigma = \frac{3FL}{2bd^2} = \frac{3 \times 960 \times 0.7}{2 \times 0.04 \times 0.015^2} = 112 \text{ MPa}$$

$$112 \text{ MPa} < 515 \text{ MPa}$$

This Specimen will not plastically deform